

Module 12 : Two-factor randomized experiments

Marie-Abèle Bind

Design of Experiments - Stat 140

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Two-factor randomized experiments (2^2 factorial design)

- Factor : variable that can be modified (i.e., intervention).
- Why varying two factors at a time ?
- Somewhat difficult, why ?

Our setting

- Let W_i denote the bivariate vector (W_{i,F_1}, W_{i,F_2}) .
- There are 2^2 treatment combinations.

$N_{(-1,-1)}$	$N_{(-1,+1)}$
$N_{(+1,-1)}$	$N_{(+1,+1)}$

Assignment mechanism

- $P(W_i=(-1,-1))=\frac{N_{(-1,-1)}}{N}$ [and so on..]

- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y}(0),\mathbf{Y}(1)) = \begin{cases} \binom{N}{N_{(-1,-1)}}^{-1} \binom{N-N_{(-1,-1)}}{N_{(-1,+1)}}^{-1} \binom{N-N_{(-1,-1)}-N_{(-1,+1)}}{N_{(+1,-1)}}^{-1} \\ \text{for } \mathbf{W} \in \mathbb{W}^+ \\ 0 \text{ otherwise} \end{cases}$$

where

$$\mathbb{W}^+ = \left\{ \mathbf{W} \text{ such that } \sum_{i=1}^N \mathbb{1}[W_i = (-1, -1)] = N_{(-1,-1)} \text{ [and so on..]} \right\}$$

- Let $Y_i(-1, -1)$, $Y_i(-1, +1)$, $Y_i(+1, -1)$, and $Y_i(+1, +1)$ denote the i^{th} unit's potential outcomes.

Unit i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$Y_1(-1, -1)$	$Y_1(-1, +1)$	$Y_1(+1, -1)$	$Y_1(+1, +1)$
2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Causal estimands

Science Table

- Let $Y_i(-1, -1)$, $Y_i(-1, +1)$, $Y_i(+1, -1)$, and $Y_i(+1, +1)$ denote the i^{th} unit's potential outcomes.

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2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Causal estimands
 - Unit-level causal effects of factor 1 treatment versus factor 1 control :

$$\tau_{i,F_1} = \frac{1}{2}[Y_i(+1, -1) - Y_i(-1, -1) + Y_i(+1, +1) - Y_i(-1, +1)]$$

Science Table

- Let $Y_i(-1, -1)$, $Y_i(-1, +1)$, $Y_i(+1, -1)$, and $Y_i(+1, +1)$ denote the i^{th} unit's potential outcomes.

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2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Causal estimands
 - Unit-level causal effects of factor 2 treatment versus factor 2 control :

$$\tau_{i,F_2} = \frac{1}{2}[Y_i(-1, +1) - Y_i(-1, -1) + Y_i(+1, +1) - Y_i(+1, -1)]$$

Science Table

- Let $Y_i(-1, -1)$, $Y_i(-1, +1)$, $Y_i(+1, -1)$, and $Y_i(+1, +1)$ denote the i^{th} unit's potential outcomes.

Unit i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$Y_1(-1, -1)$	$Y_1(-1, +1)$	$Y_1(+1, -1)$	$Y_1(+1, +1)$
2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Causal estimands
 - Average causal effect of factor 1 treatment versus factor 1 control :

$$\tau_{F_1} = \frac{1}{N} \sum_{i=1}^N \tau_{i,F_1}$$

Science Table

- Let $Y_i(-1, -1)$, $Y_i(-1, +1)$, $Y_i(+1, -1)$, and $Y_i(+1, +1)$ denote the i^{th} unit's potential outcomes.

Unit i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$Y_1(-1, -1)$	$Y_1(-1, +1)$	$Y_1(+1, -1)$	$Y_1(+1, +1)$
2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Causal estimands
 - Average causal effect of factor 2 treatment versus factor 2 control :

$$\tau_{F_2} = \frac{1}{N} \sum_{i=1}^N \tau_{i,F_2}$$

Conditional main effects

- Unit-level conditional effects of factor 1 treatment versus factor 1 control holding factor 2 at level -1 :

$$\tau_{i,F_1|F_2=-1} = [Y_i(+1, -1) - Y_i(-1, -1)]$$

Conditional main effects

- Unit-level conditional effects of factor 1 treatment versus factor 1 control holding factor 2 at level -1 :

$$\tau_{i,F_1|F_2=-1} = [Y_i(+1, -1) - Y_i(-1, -1)]$$

- Unit-level conditional effects of factor 1 treatment versus factor 1 control holding factor 2 at level +1 :

$$\tau_{i,F_1|F_2=+1} = [Y_i(+1, +1) - Y_i(-1, +1)]$$

Conditional main effects

- Unit-level conditional effects of factor 1 treatment versus factor 1 control holding factor 2 at level -1 :

$$\tau_{i,F_1|F_2=-1} = [Y_i(+1, -1) - Y_i(-1, -1)]$$

- Unit-level conditional effects of factor 1 treatment versus factor 1 control holding factor 2 at level +1 :

$$\tau_{i,F_1|F_2=+1} = [Y_i(+1, +1) - Y_i(-1, +1)]$$

- Unit-level causal effects of factor 1 treatment versus factor 1 control :

$$\tau_{i,F_1} = \frac{1}{2}[\tau_{i,F_1|F_2=-1} + \tau_{i,F_1|F_2=+1}]$$

- Unit-level causal interaction between factor 1 and factor 2 :

$$\begin{aligned}\tau_{i,F_1*F_2} &= \frac{1}{2}\{[Y_i(+1, +1) - Y_i(-1, +1)] - [Y_i(+1, -1) - Y_i(-1, -1)]\} \\ &= \frac{1}{2}[\tau_{i,F_1|F_2=+1} - \tau_{i,F_1|F_2=-1}]\end{aligned}$$

- Unit-level causal interaction between factor 1 and factor 2 :

$$\begin{aligned}\tau_{i,F_1*F_2} &= \frac{1}{2}\{[Y_i(+1,+1) - Y_i(-1,+1)] - [Y_i(+1,-1) - Y_i(-1,-1)]\} \\ &= \frac{1}{2}[\tau_{i,F_1|F_2=+1} - \tau_{i,F_1|F_2=-1}]\end{aligned}$$

- Average causal interaction between factor 1 and factor 2 :

$$\tau_{F_1*F_2} = \frac{1}{N} \sum_{i=1}^N \tau_{i,F_1*F_2}$$

Interaction

- Unit-level causal interaction between factor 1 and factor 2 :

$$\begin{aligned}\tau_{i,F_1*F_2} &= \frac{1}{2}\{[Y_i(+1,+1) - Y_i(-1,+1)] - [Y_i(+1,-1) - Y_i(-1,-1)]\} \\ &= \frac{1}{2}[\tau_{i,F_1|F_2=+1} - \tau_{i,F_1|F_2=-1}]\end{aligned}$$

- Average causal interaction between factor 1 and factor 2 :

$$\tau_{F_1*F_2} = \frac{1}{N} \sum_{i=1}^N \tau_{i,F_1*F_2}$$

- Note that, for each unit i , only one of the four potential outcomes can be observed.

Observed Table

Unit i	W_i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$(-1, -1)$	$Y_1(-1, -1)$	?	?	?
2	$(+1, -1)$	?	?	$Y_2(+1, -1)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	$(-1, +1)$	?	$Y_N(-1, +1)$	?	?

- How to write Y_i^{obs} ?

Observed Table

Unit i	W_i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$(-1, -1)$	$Y_1(-1, -1)$	?	?	?
2	$(+1, -1)$	?	?	$Y_2(+1, -1)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	$(-1, +1)$	?	$Y_N(-1, +1)$	?	?

- How to write Y_i^{obs} ?

$$Y_i^{obs} = \sum_w \mathbb{1}(W_i = w) Y_i(W_i = w)$$

Fisherian inference

Unit i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$Y_1(-1, -1)$	$Y_1(-1, +1)$	$Y_1(+1, -1)$	$Y_1(+1, +1)$
2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Sharp null hypothesis

Fisherian inference

Unit i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$Y_1(-1, -1)$	$Y_1(-1, +1)$	$Y_1(+1, -1)$	$Y_1(+1, +1)$
2	$Y_2(-1, -1)$	$Y_2(-1, +1)$	$Y_2(+1, -1)$	$Y_2(+1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1)$	$Y_N(-1, +1)$	$Y_N(+1, -1)$	$Y_N(+1, +1)$

- Test statistic

Neymanian inference

- Now suppose the observed outcomes arise from a balanced (equal number of units assigned to each treatment) 2^2 factorial design with each treatment combination assigned to $\frac{N}{4}$.

Unit i	W_i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$(-1, -1)$	$Y_1(-1, -1)$	?	?	?
2	$(+1, -1)$	?	?	$Y_2(+1, -1)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	$(-1, +1)$	?	$Y_N(-1, +1)$	?	?

- Estimator for τ_{i, F_1} ?
- Estimator for $\tau_{i, F_1 | F_2 = -1}$?
- Estimator for $\tau_{i, F_1 * F_2}$?

Neymanian inference

Unit i	W_i	$Y_i(-1, -1)$	$Y_i(-1, +1)$	$Y_i(+1, -1)$	$Y_i(+1, +1)$
1	$(-1, -1)$	$Y_1(-1, -1)$	?	?	?
2	$(+1, -1)$	?	?	$Y_2(+1, -1)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	$(-1, +1)$	?	$Y_N(-1, +1)$	?	?

- Estimator for τ_{F_1} ?
- Estimator for $\tau_{F_1|F_2=-1}$?
- Estimator for $\tau_{F_1 * F_2}$?

Neymanian inference

- $\hat{\tau}_{F_1} = \frac{1}{2}[\overline{Y}_{(+1,-1)}^{obs} - \overline{Y}_{(-1,-1)}^{obs} + \overline{Y}_{(+1,+1)}^{obs} - \overline{Y}_{(-1,+1)}^{obs}]$

Neymanian inference

- $\hat{\tau}_{F_1} = \frac{1}{2}[\overline{Y}_{(+1,-1)}^{obs} - \overline{Y}_{(-1,-1)}^{obs} + \overline{Y}_{(+1,+1)}^{obs} - \overline{Y}_{(-1,+1)}^{obs}]$
- Sampling variance of $\hat{\tau}_{F_1}$:

Neymanian inference

- $\hat{\tau}_{F_1} = \frac{1}{2}[\bar{Y}_{(+1,-1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(-1,+1)}^{obs}]$
- Sampling variance of $\hat{\tau}_{F_1}$:
 - $Var(\hat{\tau}_{F_1}) = \frac{1}{N}[S_{(-1,-1)}^2 + S_{(-1,+1)}^2 + S_{(+1,-1)}^2 + S_{(+1,+1)}^2] - \frac{1}{N}S_{F_1}^2$
where $S_{F_1}^2 = \frac{1}{N-1} \sum_{i=1}^N [\tau_{i,F_1} - \tau_{F_1}]^2$

Neymanian inference

- $\hat{\tau}_{F_1} = \frac{1}{2}[\overline{Y}_{(+1,-1)}^{obs} - \overline{Y}_{(-1,-1)}^{obs} + \overline{Y}_{(+1,+1)}^{obs} - \overline{Y}_{(-1,+1)}^{obs}]$
- Sampling variance of $\hat{\tau}_{F_1}$:
 - $Var(\hat{\tau}_{F_1}) = \frac{1}{N}[S_{(-1,-1)}^2 + S_{(-1,+1)}^2 + S_{(+1,-1)}^2 + S_{(+1,+1)}^2] - \frac{1}{N}S_{F_1}^2$
where $S_{F_1}^2 = \frac{1}{N-1} \sum_{i=1}^N [\tau_{i,F_1} - \tau_{F_1}]^2$
- Estimator for the sampling variance of $\hat{\tau}_{F_1}$:

Neymanian inference

- $\hat{\tau}_{F_1} = \frac{1}{2}[\overline{Y}_{(+1,-1)}^{obs} - \overline{Y}_{(-1,-1)}^{obs} + \overline{Y}_{(+1,+1)}^{obs} - \overline{Y}_{(-1,+1)}^{obs}]$
- Sampling variance of $\hat{\tau}_{F_1}$:
 - $Var(\hat{\tau}_{F_1}) = \frac{1}{N}[S_{(-1,-1)}^2 + S_{(-1,+1)}^2 + S_{(+1,-1)}^2 + S_{(+1,+1)}^2] - \frac{1}{N}S_{F_1}^2$
where $S_{F_1}^2 = \frac{1}{N-1} \sum_{i=1}^N [\tau_{i,F_1} - \tau_{F_1}]^2$
- Estimator for the sampling variance of $\hat{\tau}_{F_1}$:

Neymanian inference

- $\tau_{\hat{F}_1} = \frac{1}{2}[\bar{Y}_{(+1,-1)}^{obs} - \bar{Y}_{(-1,-1)}^{obs} + \bar{Y}_{(+1,+1)}^{obs} - \bar{Y}_{(-1,+1)}^{obs}]$
- Sampling variance of $\tau_{\hat{F}_1}$:
 - $Var(\tau_{\hat{F}_1}) = \frac{1}{N}[S_{(-1,-1)}^2 + S_{(-1,+1)}^2 + S_{(+1,-1)}^2 + S_{(+1,+1)}^2] - \frac{1}{N}S_{F_1}^2$
where $S_{F_1}^2 = \frac{1}{N-1} \sum_{i=1}^N [\tau_{i,F_1} - \tau_{F_1}]^2$
- Estimator for the sampling variance of $\tau_{\hat{F}_1}$:
 - $\hat{Var}(\tau_{\hat{F}_1}) = \frac{4\hat{S}^2}{N}$
where $\hat{S}^2 = \frac{1}{4}[s_{(-1,-1)}^2 + s_{(-1,+1)}^2 + s_{(+1,-1)}^2 + s_{(+1,+1)}^2]$
e.g., $s_{(-1,-1)}^2 = \frac{1}{\frac{N}{4}-1} \sum_{i: W_i=(-1,-1)} [Y_i^{obs} - \bar{Y}_{(-1,-1)}^{obs}]^2$
[Unbiased assuming $S_{(-1,-1)}^2 = S_{(-1,+1)}^2 = S_{(+1,-1)}^2 = S_{(+1,+1)}^2 = S^2$
and $S_{F_1}^2 = 0$]